



MADHUMITHA SWAMINATHAN RANGARAJAN

(BigData,Python,Spark,ETL/ELT,RDBMS,Azure,SQL)

Senior Data & Product Leader | Business

Intelligence (13+ years of experience)

Certified SAFe 6 Product Owner/ Product Manager

Microsoft Certified Azure Data Engineer

German Opportunity Card Visa | Ready to relocate

German (A1)

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[Rangarajan | LinkedIn](#)

Patent Grant with USPTO: 5

Portfolio: <https://madhu90m.github.io/portfolio-website/>

JOB SKILLS

- Adept professional with extensive experience in **Bigdata, orchestration tools, Azure, SQL, Python**
- Expertise in defining product roadmaps ensuring user experience excellence in digital products
- Collaborated with **Agile scrum** cross-functional teams to ensure on-time project deliverables
- Spearheaded **SPARK3 upgrade** program, reducing operational costs by **15%(~\$250k/year)** annually
- Led **Vertica-to-Hadoop** migration via **Sqoop, Hive, and HDFS** optimizing extraction with partitioning and predicate pushdown to cut processing time by **30%**
- Mentored **7+ junior engineers** in Data quality design approach for delivery perspective
- Built **100+ PySpark**-based data quality checks, ensuring **99%** accuracy in production
- Developed **Autosys** JIL validator tool, saving **1500+** hours(**1FTE**) and boosting cross-team efficiency
- Architected **DataStage v11.3** solution on **Unix** with detailed Statement of Work (SOW)
- Facilitated **15+** Design Thinking sessions resulting in the filing of over **20+ patents** with the **USPTO**
- Holds a **Patent Grant** with USPTO and **filed 9 patents & 5 granted** so far
- Led technical recruitment across **5+** data engineering domains, hiring **20+** skilled professionals
- Enabled GDPR-compliant data pipelines using python, boosting audit readiness and reducing risk.

TECH. SKILLS

Programming Languages: Python(ETL workflows), SQL, Pyspark, Unix shell scripting

Big Data Technologies: Apache Spark(PySpark, Spark SQL) ,HDFS ,Hive, Impala, Sqoop

ETL/ELT: Azure Synapse Analytics, Azure Data Factory, Datastage (v7.5- v11.7)

Databases: MySQL, PostgreSQL, Oracle, SQL Server, DB2, Teradata, Vertica

Cloud Platform: Microsoft Azure (Synapse, Data Factory, Databricks, Storage, Key Vault, Logic Apps, Data Explorer), Snowflake

Version Control/ Repo: GitHub Actions CI/CD, Jenkins, Bit bucket, SVN

Tools & Dataformats: JIRA, ServiceNow, Confluence, Parquet, JSON, Avro, ORC, CSV

Schedulers/Orchestration: Autosys, Control-M. Transferable to Orchestration tools

Project Management: SAFe 6 POPM, Technical Documentation, Client Communication

Compliance/ Data Governance: Data Lineage, Data classification, Azure RBAC, Key vault

Methodologies: Agile (Scrum/Kanban) , Waterfall, ITIL Process Management

AI Tools: Github Copilot Automation , GenAI

WORK EXPERIENCE

July 2019 till date

Project Management/Lead Data Engineer

Bank Of America Continuum : Chennai, India

- Led and mentored a **7-member** team, managing end-to-end delivery of scalable data solutions
- Managed Agile backlogs (**JIRA**), aligning data builds with business KPIs in Agile sprints
- Migrated from **Vertica to Hadoop (Pyspark, Sqoop, Hive, HDFS)**, cutting processing time by **30%**
- Fine-tuned complex **Vertica SQL queries** for large-scale extracts to improve retrieval speed
- Implemented **20+** reusable **Spark SQL** modules to streamline **ETL** data delivery across platform
- Documented **HLD/LLD** and architecture for **Azure Data Lake & Synapse** projects
- Delivered **Spark 3** upgrade in just **2 months** via sprint led execution
- Automated incremental data extraction from **Vertica using Sqoop** with **dynamic partitioning** logic, reducing daily load windows and minimizing impact on production systems
- Automated QA checks via **Autosys JIL** validator, saving **1500+** engineering hours
- Integrated solutions with **GitHub** Actions and **Jenkins** for **CI/CD** and deployment consistency
- **Championed performance optimization initiatives**, reducing **DataStage job runtime by 50%**, improving system efficiency and enhancing SLAs

Achievements:

Aug 2018 - Jun 2019	- Recognized as Delivery Head Innovation champion; Proposed novel ideas on Rewards Balance Prediction algorithm in Global Innovation events which is adopted org wide - Honored as a Top Women Inventor in India for filing 3 patents with USPTO on a single idea
Dec 2017 - Jul 2018	Module Lead (Data Strategy & Insights) L&T InfoTech Ltd: Chennai, India - Module Lead for a team of 5, overseeing the migration from Cognos Data Manager to Datastage v11.5 on Unix OS achieving 100% on-schedule delivery and zero critical issues post-migration - Architected migration proposals using DataStage, Unix scripting, and Control-M; improved delivery efficiency by 20% via Agile Scrum - Delivered solution docs (Design Specs, Data Mapping, Technical Workflows) for 10+ stakeholders - Validated data with 100+ complex Oracle & PostgreSQL queries to maintain 99.9% data accuracy - Enhanced DataStage jobs for 100% warning-free execution cutting maintenance overhead by 15% - Collaborated using JIRA and Bitbucket for sprint tracking and version control Achievements: - Reduced batch runtime from 3 hours to 20 minutes by implementing a data retention utility
Dec 2015 - Nov 2017	Career break (Maternity Leave)
Nov 2011 - Nov 2015	Design Lead (Data Insights) HCL Technologies: Chennai, India - Design Lead working within the Waterfall model for enhancing Risk Data Warehouse - Led Requirement Gathering from 5+ key stakeholders, translated business rules into ETL specs - Designed ETL solutions in DataStage, integrating Oracle, Control-M, and Unix Shell Scripts - Proposed data mappings & aggregation logic for centre-level financial KPIs using Oracle PL/SQL Achievements: - Implemented a Test Validation Framework using Oracle & SQL queries to ensure data accuracy - Closed audit gap with Unix Shell-SQL automation, saving 1000+ hours in batch operations
EDUCATION (May 2007- Jul 2011)	Developer/Module lead Tata Consultancy Services, India - Migrated DataStage v7.5 to v8.1; automated validation via shell scripts, cutting runtimes by 50% - Automated dataset cleanup using Shell scripts, reducing SLA breaches & improving data lifecycle Achievements: - Recognized as a “Problem Solver” for restoring critical applications during outage by leadership
LANGUAGES	Bachelor of Engineering- Electrical and Electronics Engineering Anna University Coimbatore
CERTIFICATIONS	English – C1 (Fluent) ; German – A1 certificate completed from Let’s speak Academy.
PATENTS GRANT	- Microsoft Azure Data Engineer Associate - Certified IBM Solution Developer – Infosphere Datastage v8.5 - Certified SAFe® 6 Product Owner/ Product Manager
PATENTS FILED	Madhumitha Swaminathan Rangarajan Inventions, Patents and Patent Applications - Justia Patents Search
PERSONAL SKILLS	9 Patents filed with USPTO & 5 are Granted - Passionate about building world-class digital experiences that transform industries. - Successfully led and managed cross-functional teams with diverse skill sets - Excellent communicator and team player, fostering collaboration and innovation.